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1  ///! Conditional Statements :
2  /*
3      ~ Conditional statements are used to perform different actions based on different conditions.
4      # we have several types of conditional statements, including:
5      1. if statement
6          // The if statement is used to execute a block of code if a specified condition is true.
7          // If the condition is false, the block of code will be skipped.
8          $ syntax:
9              if (condition)
10                 {
11                     code to be executed if the condition is true
12                 }
13          % note :
14              - The condition is an expression that evaluates to true or false.
15              - The code block inside the if statement will only be executed if the condition is true.
16          ////////////////////////////////////

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17
18      2. if-else statement
19          // The if-else statement is used to execute one block of code if a specified condition is true,
20          // and another block of code if the condition is false.
21          $ syntax:
22              if (condition)
23              {
24                  code to be executed if the condition is true
25              }
26              else
27              {
28                  code to be executed if the condition is false
29              }
30          % note :
31              - The condition is an expression that evaluates to true or false.
32              - The code block inside the if statement will only be executed if the condition is true,
33              while the code block inside the else statement will only be executed if the condition is false.
34          # it implement one of the block of code based on the condition, and it will not execute
35          # both blocks of code at the same time.
36          ////////////////////////////////////
37

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38      3. if-else-if statement
39          // The if-else-if statement is used to specify a new condition to test,
40          // if the first condition is false.
41          // It allows you to test multiple conditions and execute different blocks
42          // of code based on the results
43          $ syntax:
44              if (condition1)
45              {
46                  code to be executed if condition1 is true
47              }
48              else if (condition2)
49              {
50                  code to be executed if condition2 is true
51              }
52              else
53              {
54                  code to be executed if both condition1 and condition2 are false
55              }
56          # note :
57              - The conditions are expressions that evaluate to true or false.
58              - The code block inside the if statement will only be executed if condition1 is true,
59              while the code block inside the else if statement will only be executed if condition2 is true,
60              and the code block inside the else statement will only be executed if both condition1 and condition2 are false.
61          # it implement one of the block of code based on the conditions,
62          # and it will not execute more than one block of code at the same time.

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63      ! you can have as many else if statements as you need to test multiple conditions,
64      | but only one block of code will be executed based on the first true condition.
65      ///////////////////////////////////////////////////
66
67  4. switch statement
68      // The switch statement is used to perform different actions based on different conditions.
69      // It is often used as a more efficient way to compare a variable against multiple values.
70      $ syntax:
71      switch (expression)
72      {
73          case constant1:
74              code to be executed if expression equals constant1;
75              break;
76          case constant2:
77              code to be executed if expression equals constant2;
78              break;
79          ...
80          default:
81              code to be executed if expression doesn't match any constant;
82      }
83
84  */
85

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86
87  ///////////////////////////////////
88
89  //! Enums with IF Statements
90
91
92  // # example 1 : Using if statement with enum
93
94  #include <iostream>
95  using namespace std;
96
97  enum E_ScreenColor
98  {
99      RED = 1,
100     GREEN = 2,
101     BLUE = 3 ,
102     YELLOW = 4
103 };
104
105

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106 int main()
107 {
108     E_ScreenColor color;
109     int choice;
110
111     cout << "*****"<< endl;
112     cout << "please choose the number of your color ?"<< endl;
113     cout << "(1) RED" << endl;
114     cout << "(2) GREEN" << endl;
115     cout << "(3) BLUE" << endl;
116     cout << "(4) YELLOW" << endl;
117     cout << "*****"<< endl;
118     cout << "Enter your choice (1-4): "<< endl;
119
120     cin >> choice;
121     // here we are casting the integer input to the enum type E_ScreenColor
122     color = (E_ScreenColor) choice;
123
124     if (choice == E_ScreenColor::RED)
125         system("color 4F");
126     else if (choice == E_ScreenColor::GREEN)
127         system("color 2F");
128     else if (choice == E_ScreenColor::BLUE)
129         system("color 1F");
130     else if (choice == E_ScreenColor::YELLOW)
131         system("color 6F");

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132     else
133         cout << "Invalid choice!" << endl;
134
135
136     return 0;
137 }
138
139
140 //# example 2 : Using if statement with enum
141
142 enum En_countryChoice
143 {
144     JORDAN = 0,
145     USA = 1,
146     UK = 2,
147     CANADA = 3 ,
148     AUSTRALIA = 4 ,
149     OTHER = 5
150 };
151
152 int main()
153 {
154     En_countryChoice country;
155     int choice;
156

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157     cout << "*****"<< endl;
158     cout << "please choose the number of your country ?"<< endl;
159     cout << "(0) JORDAN" << endl;
160     cout << "(1) USA" << endl;
161     cout << "(2) UK" << endl;
162     cout << "(3) CANADA" << endl;
163     cout << "(4) AUSTRALIA" << endl;
164     cout << "(5) OTHER" << endl;
165     cout << "*****"<< endl;
166     cout << "Enter your choice (0-5): "<< endl;
167
168     // user input to select a country
169     cin >> choice;
170     // here we are casting the integer input to the enum type En_countryChoice
171     country = (En_countryChoice) choice;
172

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172     You, 2 weeks ago + >>>>
173     if (choice == En_countryChoice::JORDAN)
174     |     cout << "You have selected JORDAN." << endl;
175     | else if (choice == En_countryChoice::USA)
176     |     cout << "You have selected USA." << endl;
177     | else if (choice == En_countryChoice::UK)
178     |     cout << "You have selected UK." << endl;
179     | else if (choice == En_countryChoice::CANADA)
180     |     cout << "You have selected CANADA." << endl;
181     | else if (choice == En_countryChoice::AUSTRALIA)
182     |     cout << "You have selected AUSTRALIA." << endl;
183     | else if (choice == En_countryChoice::OTHER)
184     |     cout << "You have selected OTHER." << endl;
185     | else
186     |     cout << "Invalid choice!" << endl;
187
188
189     return 0;
190 }
191
192 ///////////////////////////////////////////////////
193

```

```

194 //! Switch .. Case Statement
195
196 //~ using with comparing values
197
198 //$ syntax :
199
200 switch (expression)
201 {
202     case constant1:
203         // code to be executed if expression equals constant1;
204         break;
205     case constant2:
206         // code to be executed if expression equals constant2;
207         break;
208     ...
209     default:
210         // code to be executed if expression doesn't match any constant;
211 }
212

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213  /** explain each part of the syntax:
214  /*
215      * 1. switch (expression):
216          ? This is the main part of the switch statement.
217          ? The expression is evaluated once, and its value is compared with the values of each case.
218      * 2. case constant1:
219          ? This is a case label. If the value of the expression matches constant1,
220          ? the code block following this case label will be executed.
221      * 3. break;
222          ? This statement is used to exit the switch statement.
223          ! If break is not used, the program will continue to execute the next case's code block (this is called "fall-through").
224      * 4. default:
225          ? This is an optional case that will be executed if none of the other case labels match the value of the expression.
226          ? It acts as a catch-all for any unmatched cases.
227  */
228
229  /** example 1 : Using switch statement
230

```

```

231  int main ()
232  {
233      int day = 4;
234      switch (day)
235      {
236          case 1:
237              cout << "Monday" << endl;
238              break;
239          case 2:
240              cout << "Tuesday" << endl;
241              break;
242          case 3:
243              cout << "Wednesday" << endl;
244              break;
245          case 4:
246              cout << "Thursday" << endl;
247              break;
248          case 5:
249              cout << "Friday" << endl;
250              break;
251          case 6:
252              cout << "Saturday" << endl;
253              break;
254          case 7:
255              cout << "Sunday" << endl;
256              break;
257          default:

```

```

257         default:      You, 13 hours ago • Uncommitted changes
258             cout << "Invalid day!" << endl;
259     }
260
261 }
262
263 ///////////////////////////////////////////////////
264 /// example 2 : Using switch statement with enum
265
266 enum E_WeekDays
267 {
268     MONDAY = 1,
269     TUESDAY = 2,
270     WEDNESDAY = 3 ,
271     THURSDAY = 4,
272     FRIDAY = 5,
273     SATURDAY = 6,
274     SUNDAY = 7
275 };
276
277

```

```

278 int main ()
279 {
280     E_WeekDays day;
281     int choice;
282
283     cout << "*****"<< endl;
284     cout << "please choose the number of your day ?"<< endl;
285     cout << "(1) MONDAY" << endl;
286     cout << "(2) TUESDAY" << endl;
287     cout << "(3) WEDNESDAY" << endl;
288     cout << "(4) THURSDAY" << endl;
289     cout << "(5) FRIDAY" << endl;
290     cout << "(6) SATURDAY" << endl;
291     cout << "(7) SUNDAY" << endl;
292     cout << "*****"<< endl;
293     cout << "Enter your choice (1-7): "<< endl;
294
295     cin >> choice;
296     // here we are casting the integer input to the enum type E_WeekDays
297     day = (E_WeekDays) choice;
298

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```
298 // 1 day, 42 hours ago - Uncommitted changes
299     switch (day)
300     {
301         case E_WeekDays::MONDAY:
302             cout << "You have selected MONDAY." << endl;
303             break;
304         case E_WeekDays::TUESDAY:
305             cout << "You have selected TUESDAY." << endl;
306             break;
307         case E_WeekDays::WEDNESDAY:
308             cout << "You have selected WEDNESDAY." << endl;
309             break;
310         case E_WeekDays::THURSDAY:
311             cout << "You have selected THURSDAY." << endl;
312             break;
313         case E_WeekDays::FRIDAY:
314             cout << "You have selected FRIDAY." << endl;
315             break;
316         case E_WeekDays::SATURDAY:
317             cout << "You have selected SATURDAY." << endl;
318             break;
319         case E_WeekDays::SUNDAY:
320             cout << "You have selected SUNDAY." << endl;
321             break;
322         default:
323             cout << "Invalid choice!" << endl;
324     }
325 }
326
```